The type of proofs that a type is a mere proposition, and the type of proofs that it is a set, are themselves mere propositions

The Univalent Foundations Program

hott-book:logic.tex:local-snapshot-20261012; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's labelled lemma at source lines 334-336, which carries two labels, thm:isprop-isprop and thm:isprop-isset; the proof environment starts on line 337, the line immediately after the statement environment ends, so the pairing is by adjacency and the retained proof range 337-346 is the whole environment.

Editorial note. The one-line context span at source line 332 is the author's own lead-in for the lemma and is retained so that the reader sees the lemma in its place; it carries no counter and changes no printed number.

Editorial note. The retained counters make the counted lemma print as Lemma 3.3.5, matching the original book numbering; an earlier card from this file used the same chapter and section counters with a different theorem offset.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-source copy at sources/hott-logic/retained-preamble.tex, referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: the types of mere-proposition proofs and of sethood proofs are themselves mere propositions (author label thm:isprop-isprop, original Lemma 3.3.5).

Allowed standard prerequisites (editorial).

- Function extensionality. A homotopy between two functions may be turned into an equality of functions.. It is applied twice, at source line 338 for the first half and at source lines 344-345 for the second.
- Mere proposition and set. A type is a mere proposition when all its identity types are contractible and a set when all its identity types are mere propositions; every mere proposition is a set.. It is the level information that either witness supplies at source lines 340-341 and 343-344.
- 1-type. A type is a 1-type when all its identity types are sets.. It is the level that either witness of sethood supplies at source line 344.

Source reference. Original theorem 3.3.4: sources/hott-logic/book-labelnumbers-first-edition.tex, source lines 1863–1863. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

In particular, this implies:

Lemma 3.3.5. *For any type A, the types* $\mathsf{isProp}(A)$ *and* $\mathsf{isSet}(A)$ *are mere propositions.*

Proof. Suppose $f, g : \mathsf{isProp}(A)$. By function extensionality, to show $f = g$ it suffices to show $f(x, y) = g(x, y)$ for any $x, y : A$. But $f(x, y)$ and $g(x, y)$ are both paths in A, and hence are equal because, by either f or g, we have that A is a mere proposition, and hence by original theorem 3.3.4 is a set. Similarly, suppose $f, g : \mathsf{isSet}(A)$, which is to say that for all $a, b : A$ and $p, q : a = b$, we have $f(a, b, p, q) : p = q$ and $g(a, b, p, q) : p = q$. But by then since A is a set (by either f or g), and hence a 1-type, it follows that $f(a, b, p, q) = g(a, b, p, q)$; hence $f = g$ by function extensionality. □